



CS Club Dojo's pwn.college

Players Guide



- Platform by former CTF players and ASU students (now professors)
- Heavily modified version of CTFd
- Part of ASU CS required curriculum

The Courses — Earning Credit

We leverage the above material to run a number of courses on this platform. For the most part, these courses import the above material, though some might introduce new concepts and challenges.

CSE 365 - Fall
2025

1 Module
149 Challenges

CSE 466 - Fall
2025

11 Modules
267 Challenges

CSE 539 - Spring
2025

7 Modules
44 Challenges

CSE 598 AVR - Fall
2024

7 Modules
62 Challenges

pwn.college | 🏠 Dojos 🖥️ Workspace ? Help 💬 Chat 🔍 Search



Getting Started — Learn the Basics!

These first few dojos are designed to help you Get Started with the platform. Start here before venturing onwards!

Start Here



2 Modules
11 Challenges

Linux Luminarium



16 Modules
126 Challenges

Computing 101



7 Modules
69 Challenges

Playing With
Programs



4 Modules
116 Challenges

After completing the dojos above, dive into the Core Material below!

Core Material — Earn Your Belts!

These dojos form the official pwn.college curriculum, taking you on a curated journey through the art of hacking. As you progress and build your skills, like in a martial art, you will earn **belts** for completing dojo after dojo. We won't stop you from jumping around if you want (and have the requisite skills), but you must earn belts sequentially.

Intro to
Cybersecurity



7 Modules
183 Challenges

Program Security



7 Modules
188 Challenges

System Security



6 Modules
93 Challenges

Software
Exploitation



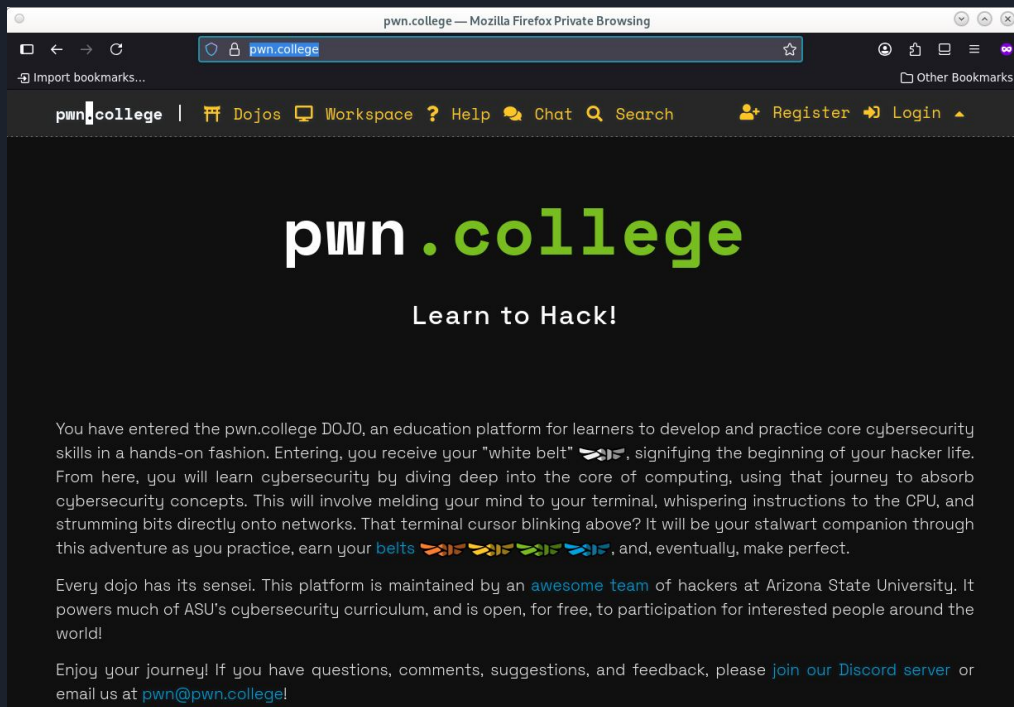
6 Modules
103 Challenges

After completing the dojos above, not only will you be added to the **belts** page, but *we will send you actual pwn.college-embroidered belts*!

To get your belt, **send us an email** from the email address associated with your pwn.college account. We'll then get your belt over to you (eventually)! Note that, due to logistical challenges, we're currently only *shipping* belts to hackers after they earn their blue belt. Until then, we will belt you in person, at ASU or some security conference.

Register

- Browse to pwn.college
- Click Register in menu bar



Ground Rules

- Educational Platform
 - Arizona State University
 - Free and Open to All
- DO NOT PUBLICLY SHARE SOLUTIONS
- Do not abuse their system



Ground Rules

When using this platform, please keep several ground rules in mind!

Writeups

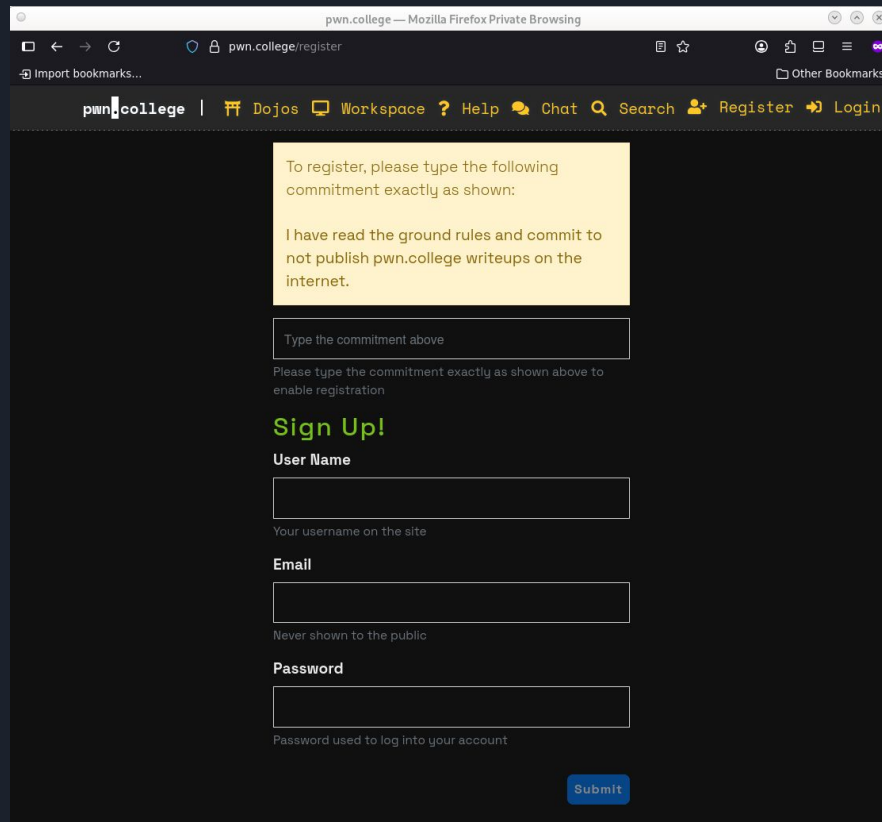
The challenges created for pwn.college are educational material, and are used to grade students at Arizona State University. Because of this, we would appreciate that writeups, walkthrough videos, and livestreams of challenge solutions are not posted to the internet. Obviously, we can't stop you from posting things to the internet, but we worked hard to make all of this public, and we would appreciate your help in keeping pwn.college a viable educational platform.

Bug Bounties

Even master hackers have vulnerabilities in their software! If you find a vulnerability in this platform, please [email](#) us rather than pwn us. We'll appreciate it, and you'll get extra credit (if you're an ASU student) or an awesome scoreboard award!

Integrity Pledge

- Agree to the rules
- Avoid using school email
 - Continue to use after graduation



The screenshot shows the registration page for pwn.college in a Mozilla Firefox Private Browser. The page has a dark theme with a yellow commitment box at the top. Below the commitment box is a text input field for the pledge, followed by a confirmation message. The registration form includes fields for User Name, Email, and Password, each with a label and a description. A blue Submit button is at the bottom right.

pwn.college — Mozilla Firefox Private Browsing

pwn.college/register

Import bookmarks... Other Bookmarks

pwn.college | Dojos Workspace ? Help Chat Search Register Login

To register, please type the following commitment exactly as shown:

I have read the ground rules and commit to not publish pwn.college writeups on the internet.

Type the commitment above

Please type the commitment exactly as shown above to enable registration

Sign Up!

User Name

Your username on the site

Email

Never shown to the public

Password

Password used to log into your account

Submit



Join Westshore Dojos

- Click join links while logged into pwn.college to join extra Dojos

Join [Learning Python \(High School\) Dojo](#)

Join [Westshore Jr/Sr High CS Club Dojo](#)

Learning Python (High School)

9 Modules
44 Challenges



Westshore Jr/Sr High CS Club

12 Modules
50 Challenges





Starting pwn.college

Start Here

29 Hacking
2 Modules
10 Challenges



Modules

Using the Dojo

29 Hacking
0 / 10

Joining the Discord

0 / 0

White Belt Challenges

- Walkthrough on how to use pwn.college
 - The video is pretty old now...

Interacting with the Dojo



Getting Started



Using the Terminal

3 hacking, 43782 solves



Using the VSCode Workspace

3 hacking, 57090 solves



Using the GUI Desktop

3 hacking, 53051 solves



Pasting into the Desktop

50278 solves



Connecting over SSH

OPTIONAL

4 hacking, 11144 solves



Restarting Challenges

3 hacking, 44329 solves



Getting Help

OPTIONAL

27259 solves

Welcome to the Dojo

- Many Dojos available to teach new skills
- Each dojo has submodules on different subjects
- Module contents
 - Youtube lecture
 - Google slides
 - Challenges
- Completing Dojos earns you rewards
- Completing Core Dojos earns you an actual pwn.college belt!



Getting Started — Learn the Basics!

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Start Here	Linux Luminarium	Computing 101	Playing With Programs
2 Modules 11 Challenges	16 Modules 127 Challenges	7 Modules 69 Challenges	4 Modules 116 Challenges

After completing the dojos above, dive into the Core Material below!

Core Material — Earn Your Belts!

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Intro to Cybersecurity	Program Security	System Security	Software Exploitation
7 Modules 163 Challenges	6 Modules 169 Challenges	6 Modules 93 Challenges	6 Modules 90 Challenges

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Community Material — Earn Badges!

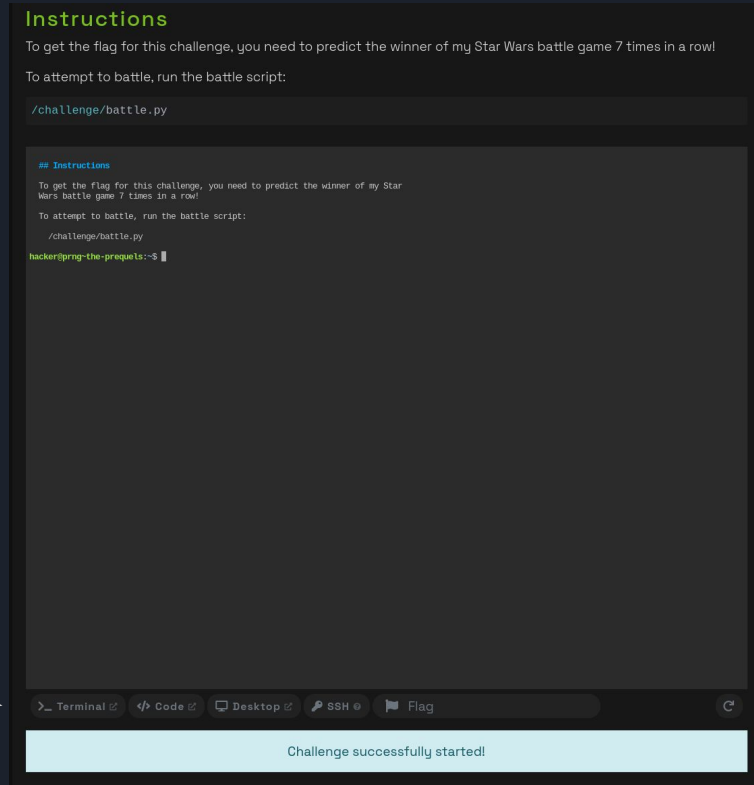
No matter how much material we create, there is always more to learn! This section contains additional dojos created by the pwn.college community. Some are designed to be tackled after you complete the dojos above, whereas others are open to anyone interested in more specialized topics.

pwn.college Archives	ACSAC 2024 CTF	ARM Architecture	CTF Archive
7 Modules 201 Challenges	1 Module 11 Challenges	2 Modules 28 Challenges	89 Modules 690 Challenges

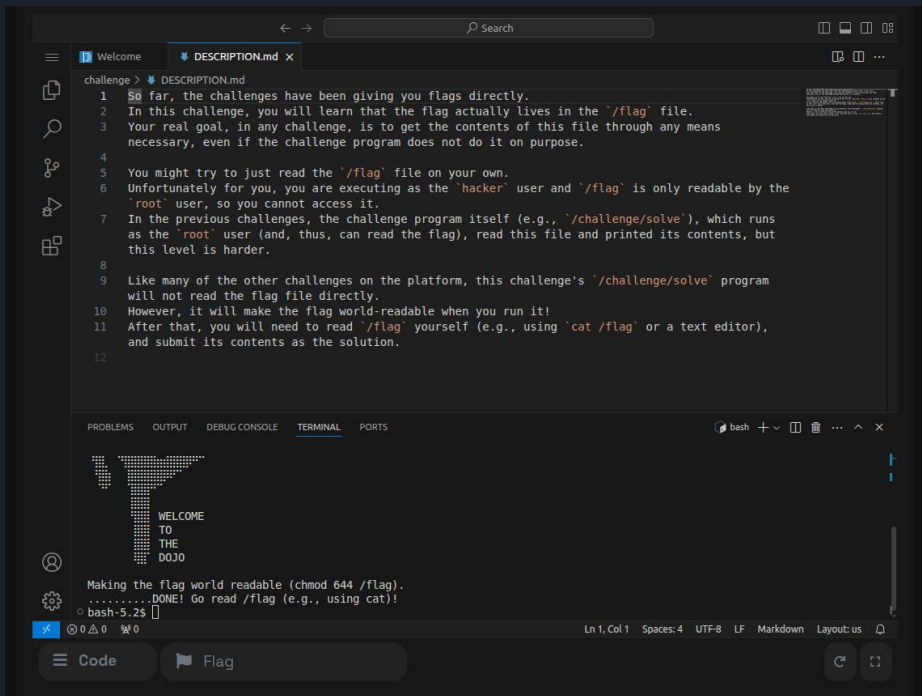
Demo dojo	Intro to Programming Languages	The Quarterly Quiz	Windows Warzone
1 Module 0 Challenges	23 Modules 219 Challenges	2 Modules 13 Challenges	2 Modules 17 Challenges

Completing Challenges

- Pwn.college hosts a VM in the cloud (ASU)
 - You don't need a powerful computer
 - Just need a web browser
 - Security and development tools already installed!
- Core challenges are designed
 - Teach security concepts
 - Demonstrate new skills
 - Small incremental steps
- Start the challenge
 - Web-terminal
 - VS Code
 - Desktop
 - SSH mode



Other options

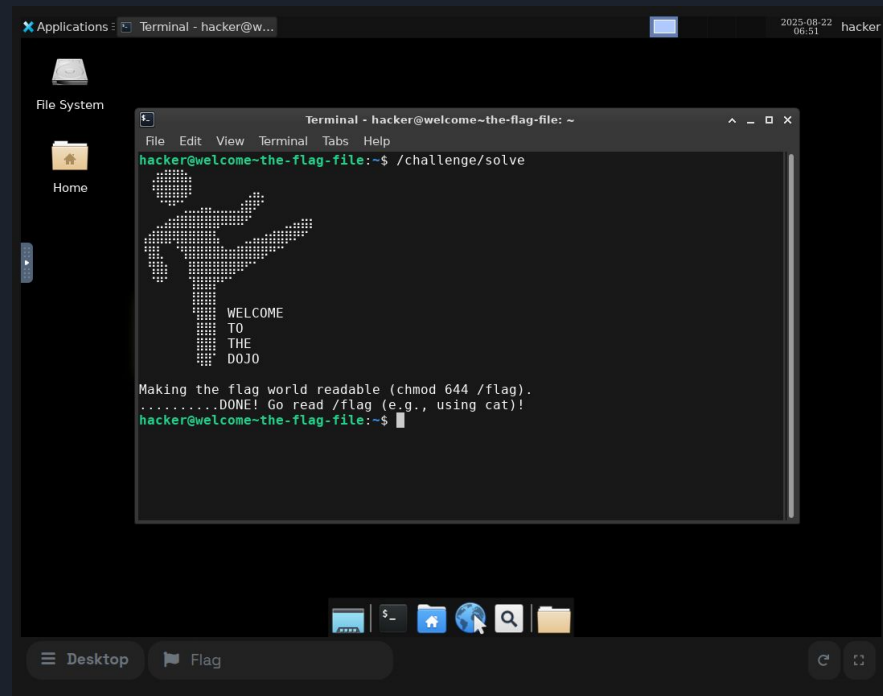


The screenshot shows a code editor with a file named `DESCRIPTION.md` open. The file contains a challenge description with 12 lines of text. Below the editor, there is a terminal window. The terminal shows a ASCII art of a person standing, followed by the text "WELCOME TO THE DOJO". Below this, it says "Making the flag world readable (chmod 644 /flag)." and ".....DONE! Go read /flag (e.g., using cat)!". The terminal prompt is `bash-5.2$`.

```
challenge > DESCRIPTION.md
1 50 far, the challenges have been giving you flags directly.
2 In this challenge, you will learn that the flag actually lives in the '/flag' file.
3 Your real goal, in any challenge, is to get the contents of this file through any means
4 necessary, even if the challenge program does not do it on purpose.
5
6 You might try to just read the '/flag' file on your own.
7 Unfortunately for you, you are executing as the 'hacker' user and '/flag' is only readable by the
8 'root' user, so you cannot access it.
9 In the previous challenges, the challenge program itself (e.g., '/challenge/solve'), which runs
10 as the 'root' user (and, thus, can read the flag), read this file and printed its contents, but
11 this level is harder.
12
13 Like many of the other challenges on the platform, this challenge's '/challenge/solve' program
14 will not read the flag file directly.
15 However, it will make the flag world-readable when you run it!
16 After that, you will need to read '/flag' yourself (e.g., using 'cat /flag' or a text editor),
17 and submit its contents as the solution.
```

```
WELCOME
TO
THE
DOJO

Making the flag world readable (chmod 644 /flag).
.....DONE! Go read /flag (e.g., using cat)!
bash-5.2$
```



The screenshot shows a terminal window titled "Terminal - hacker@w...". The terminal prompt is `hacker@welcome-the-flag-file:~$`. The user has entered the command `/challenge/solve`. The output shows a ASCII art of a person standing, followed by the text "WELCOME TO THE DOJO". Below this, it says "Making the flag world readable (chmod 644 /flag)." and ".....DONE! Go read /flag (e.g., using cat)!". The terminal prompt is `hacker@welcome-the-flag-file:~$`.

```
hacker@welcome-the-flag-file:~$ /challenge/solve

WELCOME
TO
THE
DOJO

Making the flag world readable (chmod 644 /flag).
.....DONE! Go read /flag (e.g., using cat)!
hacker@welcome-the-flag-file:~$
```



Connectivity Recommendations

- Desktop easiest for new users / user friendly
 - Not going to work well if Internet connection slow
 - You can't cut/paste flags directly, use the clipboard on side of VM
- VS Code is very powerful
 - Probably best way to write code / syntax highlighting
 - Save often! Not sure how good recovery is if you lose internet connection...
- SSH is what I use most often
 - Fastest / even if internet connection slow
 - It's what you will use most often in the real world for remote access
 - You can SSH when running challenge
 - Even while using the other GUI Desktop or VS Code
 - It's always available
 - Learn tmux!!!

pwn.college dojo order

1. [Start Here](#) - Teach how pwn.college works
2. [Linux Luminarium](#) - How to use Linux

Once you have powered up with the first 2 dojos, you can try these next dojos in any order / what interests you

- [Learning Python](#) - Python CLI Programming
- [Playing with Programs](#) - Data, HTTP, suid
- [Westshore Jr/Sr HS CS Club](#) - Beginner CTF
- [Intro to Programming Languages](#) - C / C++
- [Computing 101](#) - Assembly and GDB



Arizona State University



- Computer Science (Cybersecurity), BS - 120 credit hours

Computer Science Lower Division
CSE 110 Principles of Programming (QTRS)
EEE 120 Digital Design Fundamentals
CSE 205 Object-Oriented Programming and Data Structures (QTRS)
CSE 230 Computer Organization and Assembly Language Programming
CSE 240 Introduction to Programming Languages
FSE 100 Introduction to Engineering
Computer Science Upper Division
CSE 301 Computing Ethics
CSE 310 Data Structures and Algorithms
CSE 330 Operating Systems
CSE 340 Principles of Programming Languages
CSE 355 Introduction to Theoretical Computer Science
CSE 360 Introduction to Software Engineering
CSE 365 Information Assurance
CSE 485 Computer Science Capstone Project I
CSE 486 Computer Science Capstone Project II
MAT 343 Applied Linear Algebra
IEE 380 Probability and Statistics for Engineering Problem Solving (QTRS)

pwn.college

Plus CSE598 - Special Topics

- Fall 24-Applied Vulnerability Research
- Spring 24 - Advanced SW Exploitation

Cybersecurity Focus Courses
^ Upper Division Cybersecurity Focus Courses
CSE 466 Computer Systems Security
CSE 467 Data and Information Security
CSE 468 Computer Network Security
CSE 469 Computer and Network Forensics
CSE 494 Topic: Artificial Intelligence for Cyber Security
CSE 412 Database Management
OR CSE 434 Computer Networks
OR CSE 445 Distributed Software Development
^ Upper Division Cybersecurity Electives
CSE 445 Distributed Software Development
CSE 460 Software Analysis and Design
CSE 463 Introduction to Human Computer Interaction
CSE 464 Software Quality Assurance and Testing
CSE 466 Computer Systems Security
CSE 467 Data and Information Security
CSE 468 Computer Network Security
CSE 469 Computer and Network Forensics
CSE 471 Introduction to Artificial Intelligence
CSE 494 Topic: Artificial Intelligence for Cyber Security



Questions

My response time can be a little slow:

`mwales3 at gmail dot com`